

not concerned with the limitations of the precision in measurements done at the subatomic level. We are just dealing with billiard balls!

In a dynamical system, where you are considering more than a ball on its own, where trajectories in a way depend on one another, the ability to project into the future is not just reduced, but is subjected to a fundamental limitation. Poincaré proposed that we can only work with qualitative matters—some property of systems can be *discussed*, but not computed. You can think rigorously, but you cannot use numbers. Poincaré even invented a field for this, analysis in situ, now part of topology. Prediction and forecasting are a more complicated business than is commonly accepted, but it takes someone who knows mathematics to understand that. To accept it takes both understanding and courage.

In the 1960s the MIT meteorologist Edward Lorenz rediscovered Poincaré's results on his own—once again, by accident. He was producing a computer model of weather dynamics, and he ran a simulation that projected a weather system a few days ahead. Later he tried to repeat the same simulation with the exact same model and what he thought were the same input parameters, but he got wildly different results. He initially attributed these differences to a computer bug or a calculation error. Computers then were heavier and slower machines that bore no resemblance to what we have today, so users were severely constrained by time. Lorenz subsequently realized that the consequential divergence in his results arose not from error, but from a small rounding in the input parameters. This became known as the butterfly effect, since a butterfly moving its wings in India could cause a hurricane in New York, two years later. Lorenz's findings generated interest in the field of chaos theory.

Naturally researchers found predecessors to Lorenz's discovery, not only in the work of Poincaré, but also in that of the insightful and intuitive Jacques Hadamard, who thought of the same point around 1898, and then went on to live for almost seven more decades—he died at the age of ninety-eight.*

They Still Ignore Hayek

Popper and Poincaré's findings limit our ability to see into the future, making it a very complicated reflection of the past—if it is a reflection of the past at all. A potent application in the social world comes from a friend of Sir Karl, the intuitive economist Friedrich Hayek. Hayek is one of the rare celebrated members of his “profession” (along with J. M. Keynes and G.L.S. Shackle) to focus on true uncertainty, on the limitations of knowledge, on the unread books in Eco's library.